

VR-Numbers.

Operational Superstructures of Integers, Rationals, Reals, and Complex Numbers over the Natural Numbers of VR

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Abstract

Building on VR (Reznik, 2026, DOI: 10.5281/zenodo.20212092), a formal axiomatization of arithmetic with three primitives $\{\emptyset, \rightarrow, t\}$ and four axioms, we develop *VR-Numbers* — a program of operational superstructures yielding $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ over VR's natural numbers $\mathbb{N}$. The construction uses no ordered pairs as ontological objects; each numerical extension is defined as a formal language of expressions with equivalence relations and operations. The ontological foundation of the entire system is the single primitive $\emptyset$; all numerical objects — including $\mathbb{N}$ — arise as operational constructions of progressively greater depth. The classical actual infinity is replaced by *operational infinity*: A4 (induction) is read as an operational principle, not as a postulate of completed infinite totalities. The two-dimensionality of $\mathbb{C}$ is *structurally grounded* in the duality present in axiom A1 of VR (the two generating implications $F \rightarrow F$ and $F \rightarrow \top$), rather than postulated through pairs of reals; the algebraic coupling of the two axes requires an additional joining axiom ($i^2 = \mathbb{Z} 1$). $\mathbb{Q}_{\text{VR}}$, $\mathbb{C}_{\text{VR}}$, and $\mathbb{R}_{\text{VR}}$ are shown to be isomorphic to the corresponding standard structures; $\mathbb{R}_{\text{VR}}$ is isomorphic to the field of computable real numbers — a countable subfield of classical $\mathbb{R}$ — reflecting the operational character of the construction.

Keywords: foundations of mathematics, operational extensions, operational infinity, integer construction, rational construction, real numbers, complex numbers, Cauchy sequences, axiomatic duality, ontology of numbers, computable analysis.

Part I. Foundation: VR and $\mathbb{N}$

This work builds upon the formal system VR introduced in [Reznik 2026]. We briefly recall its essentials.

Primitives of VR.

Constant: $\emptyset$. Binary operator: $\rightarrow$ (implication). Unary operator: t (succession).

Axioms.

A1 (Generativity). $F \rightarrow F = \top$ and $F \rightarrow \top = \top$, where F is identified with $\emptyset$ at the logical level, and $\top := F \rightarrow F$.

A2 (Implication). $\rightarrow$ is the classical implication on $\{F, \top\}$.

A3 (Succession). $t(x) = x \cup \{x\}$, so $x \in t(x)$ and $x \subset t(x)$.

A4 (Induction). The objects $O_n = t^n(\emptyset)$ exhaust all objects generated from $\emptyset$ via t .

Natural Numbers.

$O_0 = \emptyset$, $O_1 = t(\emptyset) = \{\emptyset\}$, $O_2 = t(t(\emptyset)) = \{\emptyset, \{\emptyset\}\}$, ...

The natural number n is identified with the expression O_n . Equality is defined via Leibniz's principle: $x = y \iff \forall p: p(x) \leftrightarrow p(y)$. VR has been shown to be arithmetically equivalent to Peano arithmetic, and consistent relative to ZF.

Ontological Status.

In VR, the only ontological primitive is $\emptyset$. All other objects of the system — including the natural numbers themselves — arise as operational results of applying the succession operator t to $\emptyset$. The natural number O_n is not an independent existing thing; it is the result of n applications of t to $\emptyset$. The mathematical content of n is its operational history.

This ontological position is the foundation of the present work. It asserts a minimal ontology — only $\emptyset$ exists — and treats all other numerical constructions as operations of progressively greater depth performed on this single primitive.

Part II. Integers $\mathbb{Z}$ as an Operational Superstructure

We construct integers as formal expressions over $\mathbb{N}$ with prescribed equivalence and operations.

II.1. The Formal Language of Subtraction

We extend the language of VR by a binary symbol $\ominus$. An expression of the form

$$a \ominus b, \text{ where } a, b \in \mathbb{N}$$

is a syntactic object intended to denote the operation "subtract b from a ". This is not an ordered pair, but a formal record of an operation.

II.2. Equivalence Relation

Two expressions are declared equivalent when they would yield the same result, using only the addition operation already defined on $\mathbb{N}$:

$$a \ominus b \approx c \ominus d \iff a + d = b + c.$$

This relation is reflexive, symmetric, and transitive (direct verification using the commutativity and associativity of addition in $\mathbb{N}$, established in VR).

II.3. Operations

Addition.

$$(a \ominus b) \oplus (c \ominus d) := (a + c) \ominus (b + d).$$

Multiplication.

$$(a \ominus b) \otimes (c \ominus d) := (a \times c + b \times d) \ominus (a \times d + b \times c).$$

Additive Inverse.

$$\ominus(a \ominus b) := b \ominus a.$$

Subtraction.

$$(a \ominus b) \boxminus (c \ominus d) := (a + d) \ominus (b + c).$$

Remark. Subtraction is definable through addition and additive inverse: $(a \ominus b) \boxminus (c \ominus d) = (a \ominus b) \oplus \ominus(c \ominus d)$. We list it explicitly for convenience.

All operations are well-defined on equivalence classes.

II.4. Canonical Form

Every expression $a \ominus b$ can be canonized uniquely:

If $a \geq b$, the canonical form is $(a - b) \ominus 0$.

If $a < b$, the canonical form is $0 \ominus (b - a)$.

II.5. Embedding $\mathbb{N} \rightarrow \mathbb{Z}_{\text{VR}}$

The natural number $n \in \mathbb{N}$ embeds into $\mathbb{Z}_{\text{VR}}$ as the class of $n \ominus 0$.

II.6. Isomorphism Theorem

Theorem. The set $\mathbb{Z}_{\text{VR}}$ of equivalence classes of expressions $a \ominus b$ with operations $\oplus, \otimes, \boxminus$ is a commutative ring with unit, isomorphic to the standard ring of integers $\mathbb{Z}$.

Proof sketch. Define $\varphi: \mathbb{Z}_{\text{VR}} \rightarrow \mathbb{Z}$ by $\varphi([a \ominus b]) = a - b$. This map is well-defined on classes, respects operations, and is bijective.

Part III. Rationals $\mathbb{Q}$ as an Operational Superstructure

Rational numbers are constructed analogously to integers, but over $\mathbb{Z}_{\text{VR}}$.

III.1. The Formal Language of Division

We introduce a binary symbol $\oslash$. An expression of the form

$$a \oslash b, \text{ where } a, b \in \mathbb{Z_VR} \text{ and } b \neq 0_{\mathbb{Z}}$$

is a syntactic object denoting the operation "divide a by b". The restriction $b \neq 0_{\mathbb{Z}}$ is syntactic and is effectively decidable via the canonical form in $\mathbb{Z_VR}$.

III.2. Equivalence Relation

$$a \oslash b \approx_{\mathbb{Q}} c \oslash d \iff a \otimes d \approx b \otimes c,$$

where $\otimes$ is the multiplication operation in $\mathbb{Z_VR}$.

III.3. Operations

Addition.

$$(a \oslash b) \boxplus (c \oslash d) := (a \otimes d \oplus b \otimes c) \oslash (b \otimes d).$$

Multiplication.

$$(a \oslash b) \boxtimes (c \oslash d) := (a \otimes c) \oslash (b \otimes d).$$

Subtraction.

$$(a \oslash b) \boxminus (c \oslash d) := (a \otimes d \boxminus b \otimes c) \oslash (b \otimes d).$$

Division.

$$(a \oslash b) \oslash (c \oslash d) := (a \otimes d) \oslash (b \otimes c), \text{ provided } c \neq 0_{\mathbb{Z}}.$$

III.4. Canonical Form

Every fraction $a \oslash b$ can be brought to canonical form by making the denominator positive and reducing to lowest terms.

III.5. Embedding $\mathbb{Z} \rightarrow \mathbb{Q_VR}$

The integer $a \in \mathbb{Z_VR}$ embeds into $\mathbb{Q_VR}$ as the class of $a \oslash 1_{\mathbb{Z}}$, where $1_{\mathbb{Z}} = 1 \ominus 0$ is the multiplicative identity in $\mathbb{Z_VR}$.

III.6. Isomorphism Theorem

Theorem. $\mathbb{Q_VR}$ with operations $\boxplus$, $\boxtimes$, $\boxminus$, $\oslash$ is a field, isomorphic to the standard field of rationals $\mathbb{Q}$.

Part IV. Real Numbers $\mathbb{R}$ via Cauchy Sequences

Real numbers require a qualitatively different construction: while $\mathbb{Z}$ and $\mathbb{Q}$ are syntactic constructions over finite expressions, $\mathbb{R}$ requires infinite processes.

IV.1. Functions as Operational Rules

We extend the language of VR by the notion of function $\mathbb{N} \rightarrow \mathbb{Q_VR}$. A function $a: \mathbb{N} \rightarrow \mathbb{Q_VR}$ is an operational rule: a finite description (algorithm) producing, for each $n \in \mathbb{N}$, a value $a(n) \in \mathbb{Q_VR}$. Functions are not introduced as sets of ordered pairs; they are operational primitives — finite specifications of infinite correspondences.

A consequence of this definition is that each function is fully captured by its finite description. The collection of all such finite descriptions is itself operationally countable: any function in VR-Numbers can be enumerated through its description. This will have implications for the relationship between $\mathbb{R_VR}$ and the classical $\mathbb{R}$, discussed below.

Operationally, a function $\mathbb{N} \rightarrow \mathbb{Q_VR}$ is a partial recursive function in the sense of Church–Turing, or equivalently a Type-2 computable function in the framework of computable analysis (Weihrauch, 2000). The construction $\mathbb{R_VR}$ therefore coincides with the standard development of computable real numbers via computable Cauchy sequences (Pour-El & Richards, 1989). The novelty here is not technical but *ontological*: in VR-Numbers, this is not one approach among several but the only available approach, dictated by the operational ontology.

IV.2. Fundamental (Cauchy) Sequences

A function $a: \mathbb{N} \rightarrow \mathbb{Q_VR}$ is fundamental (a Cauchy sequence) if:

$$\forall \varepsilon \in \mathbb{Q_VR}, \varepsilon > 0, \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N: |a(m) \boxminus a(n)| < \varepsilon.$$

IV.3. Equivalence of Sequences

$$a \approx_C b \iff \forall \varepsilon > 0, \exists N: \forall n \geq N: |a(n) \boxminus b(n)| < \varepsilon.$$

IV.4. The Construction $\mathbb{R_VR}$

$\mathbb{R_VR}$ is defined as the set of equivalence classes of fundamental sequences with respect to $\approx_C$.

IV.5. Operations

Operations are defined componentwise on representatives:

$$[a] \oplus_{\mathbb{R}} [b] := [c], \text{ where } c(n) = a(n) \boxplus b(n);$$

$$[a] \otimes_{\mathbb{R}} [b] := [c], \text{ where } c(n) = a(n) \boxtimes b(n);$$

$$[a] \boxminus_{\mathbb{R}} [b] := [c], \text{ where } c(n) = a(n) \boxminus b(n);$$

$$[a] \oslash_{\mathbb{R}} [b] := [c], \text{ where } c(n) = a(n) \oslash b(n).$$

IV.6. Embedding $\mathbb{Q} \rightarrow \mathbb{R_VR}$

Each rational $q \in \mathbb{Q_VR}$ embeds into $\mathbb{R_VR}$ as the class of the constant sequence $a(n) = q$ for all n .

IV.7. Isomorphism Theorem

Theorem. $\mathbb{R}_{\text{VR}}$ is a real closed ordered field. It is isomorphic to the field of computable real numbers — a countable subfield of the classical real line $\mathbb{R}$ — and contains all reals that admit a finite algorithmic description.

Remark. Strictly speaking, $\mathbb{R}_{\text{VR}}$ is not isomorphic to the entire classical $\mathbb{R}$. Classical $\mathbb{R}$ has cardinality of the continuum; $\mathbb{R}_{\text{VR}}$, consisting of operational rules each given by a finite description, is countable. The relationship is therefore: $\mathbb{R}_{\text{VR}}$ is isomorphic to the computable reals, which constitute a countable real closed subfield of classical $\mathbb{R}$. All real numbers encountered in practice — algebraic numbers, rationals, π , e , and indeed any real definable by a finite description — are captured by $\mathbb{R}_{\text{VR}}$. Non-computable reals of classical $\mathbb{R}$ have no place in the operational ontology of VR-Numbers, as they correspond to no finite algorithmic description. This aligns VR-Numbers with the constructive tradition (Bishop, 1967; Martin-Löf, 1984) and remains consistent with the philosophical position that mathematical objects are operations, not transcendent entities.

IV.8. Operational Character of $\mathbb{R}$

A real number in VR is, formally, an algorithm — a finite description specifying how to compute arbitrarily close rational approximations. The square root of 2, for instance, is represented by an algorithm (such as Newton's method) that, given any precision, produces a rational approximation within that precision. This view aligns naturally with constructive mathematics.

Part V. Complex Numbers $\mathbb{C}$ via the Duality of A1

The construction of complex numbers introduces the central new conceptual element of this work. In standard mathematics, $\mathbb{C}$ is built as ordered pairs of real numbers (a, b) with a special multiplication rule, and the two-dimensionality of $\mathbb{C}$ is postulated through the choice of pairs. We propose instead that the two-dimensionality of $\mathbb{C}$ is structurally grounded in the logical foundation of VR — specifically, in the duality present in axiom A1.

V.1. The Duality of A1

Axiom A1 of VR states:

$$F \rightarrow F = T,$$

$$F \rightarrow T = T.$$

These are two distinct generating facts: F generates both values via implication. The two implications represent two distinct movements from F :

$F \rightarrow F$: a movement returning to F itself (self-reference at the logical level).

$F \rightarrow T$: a movement to the opposite value T .

We propose to interpret these two implications as two independent axes of numerical extension.

V.2. The Two Axes

The axis $F \rightarrow F$ is the real axis: along this axis, the constructions $\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R}$ developed above unfold.

The axis $F \rightarrow T$ is the imaginary axis: along this axis, new operational expressions of the form $b \cdot i$ live, where i is an operational marker indicating the imaginary axis.

The marker i is not a number nor an object — it is a syntactic indicator of which axis a value belongs to.

V.3. The Joining Rule

To make $\mathbb{C}$ a single algebraic structure rather than two disjoint copies of $\mathbb{R}$, the two axes must be coupled. This coupling is captured by the rule:

$$i \otimes i := \ominus 1.$$

This rule is postulated as an axiom of joining: applying the imaginary marker twice is equivalent to negation along the real axis. Without this rule, the two axes would remain independent; with it, they are intertwined into the algebraic structure of $\mathbb{C}$.

V.4. Complex Numbers as Operational Expressions

A complex number is a formal expression

$$a \oplus b \cdot i, \text{ where } a, b \in \mathbb{R_VR}.$$

This is not an ordered pair (a, b) . It is a syntactic record of two values along two distinct axes.

V.5. Equivalence

$$a \oplus b \cdot i \approx c \oplus d \cdot i \iff a = c \text{ (in } \mathbb{R_VR} \text{) and } b = d \text{ (in } \mathbb{R_VR} \text{)}.$$

V.6. Operations

Addition.

$$(a \oplus b \cdot i) \oplus_{\mathbb{C}} (c \oplus d \cdot i) := (a \oplus c) \oplus (b \oplus d) \cdot i.$$

Multiplication (using $i \otimes i = \ominus 1$).

$$(a \oplus b \cdot i) \otimes_{\mathbb{C}} (c \oplus d \cdot i) := (a \otimes c \boxminus b \otimes d) \oplus (a \otimes d \oplus b \otimes c) \cdot i.$$

Conjugation.

$$\text{conj}(a \oplus b \cdot i) := a \oplus (\ominus b) \cdot i.$$

Modulus.

$$|a \oplus b \cdot i| := \sqrt{(a \otimes a \oplus b \otimes b)}.$$

V.7. Embedding $\mathbb{R} \rightarrow \mathbb{C}_{\text{VR}}$

Each real $a \in \mathbb{R}_{\text{VR}}$ embeds into $\mathbb{C}_{\text{VR}}$ as the expression $a \oplus 0 \cdot i$.

V.8. Isomorphism Theorem

Theorem. $\mathbb{C}_{\text{VR}}$ with the operations defined above is a field, isomorphic to the standard field of complex numbers $\mathbb{C}$ (with $\mathbb{R}$ replaced by $\mathbb{R}_{\text{VR}}$; the resulting field is isomorphic to the field of computable complex numbers).

V.9. Discussion: Foundation of Two-Dimensionality

In the standard treatment, the dimensionality of $\mathbb{C}$ is a postulate: $\mathbb{C}$ is defined as $\mathbb{R} \times \mathbb{R}$ with a specific multiplication, and there is no deeper justification for why $\mathbb{C}$ has exactly two dimensions.

In VR-Numbers, the two-dimensionality of $\mathbb{C}$ is structurally grounded in the logical foundation of the system. Axiom A1 contains exactly two generating implications, $F \rightarrow F$ and $F \rightarrow \neg F$. These two implications correspond to two distinct axes of numerical construction. The dimensionality of $\mathbb{C}$ is therefore not arbitrary; it reflects the dual generative structure already present in the logical layer of VR.

Scope of this claim. The duality of A1 provides two distinct generative directions but does not by itself entail their algebraic coupling. The rule $i^2 = -1$ is an additional postulate joining the two axes into a single field. Thus A1 *motivates* the two-dimensionality of $\mathbb{C}$; the field structure on these two dimensions requires the joining axiom. The conceptual content of the claim is that the two-dimensionality is not arbitrary — it has a logical origin — not that the entire algebraic structure of $\mathbb{C}$ is derivable from A1 alone.

Position relative to higher-dimensional algebras. The duality of A1 yields exactly two generative directions at the logical level. The theorems of Frobenius (1878) and Hurwitz (1898) show that further normed division algebras over $\mathbb{R}$ — quaternions $\mathbb{H}$ and octonions $\mathbb{O}$ — require successive weakening of algebraic properties (commutativity, then associativity). This is consistent with the position that A1 as a logical primitive does not further subdivide: extensions beyond $\mathbb{C}$ require leaving the purely logical layer and accepting algebraic compromises. Whether such higher-dimensional algebras admit analogous derivations from deeper logical structure remains an open question (see Part VII).

This conceptual claim — that the structure of complex numbers is not arbitrary but reflects how implication operates on falsity — is the principal philosophical contribution of the present work. The two generating movements from F (one returning to F itself, one moving to $\neg F$) are precisely the two axes that, when joined by the rule $i^2 = -1$, yield the field of complex numbers.

Part VI. The Ontological Position of VR-Numbers

VR-Numbers adopts a minimal ontology: the only ontological primitive is the void $\emptyset$. All other objects — natural, integer, rational, real, and complex numbers — are operational constructions of progressively greater depth performed over $\emptyset$.

VI.1. The Single Ontological Primitive

$\emptyset$ is the unique ontological primitive of the system. It is given as Leibnizian initial datum — that which is, prior to any operation, division, or distinction. Every other object of VR or VR-Numbers is the result of operations applied to $\emptyset$.

The succession operator t is not itself an object but a rule of operation. The natural numbers $O_n = t^n(\emptyset)$ are records of successive applications of t . Their content is fully captured by their operational history: O_3 is "three applications of t to $\emptyset$ ", and this is the totality of what O_3 is.

VI.2. Depths of Operationality

All numerical systems constructed in this work are operational, but they exhibit different depths of operational character:

System	Operational Character
$\emptyset$	The single ontological primitive — that which is, prior to all operation
$\mathbb{N}$	Direct applications of t to $\emptyset$: $O_n = t^n(\emptyset)$
$\mathbb{Z}$	Finite syntactic expressions $a \ominus b$ over $\mathbb{N}$, on the $F \rightarrow F$ axis
$\mathbb{Q}$	Finite syntactic expressions $a \oslash b$ over $\mathbb{Z}$, on the $F \rightarrow F$ axis
$\mathbb{R}$	Algorithms producing infinite sequences of rationals, on the $F \rightarrow F$ axis
$\mathbb{C}$	Two-axis syntax over $F \rightarrow F$ and $F \rightarrow T$, joined by $i^2 = \ominus 1$

VI.3. Operational Infinity

The axiom A4 of VR — the principle of induction — is conventionally read as postulating actual infinity: the completed totality of all natural numbers existing as an object. We reject this reading. Within the ontological position of VR-Numbers, A4 does not assert the existence of any infinite object. It formulates an operational principle: any property holding for $\emptyset$ and preserved under application of t holds for every result of iterated application of t . The quantifier "for all n " in A4 is operational, not ontological — it expresses universal applicability of an operation, not membership in a completed set.

We call this *operational infinity*, in contrast to the classical notion of actual infinity. Operational infinity is mathematically equivalent to actual infinity: every theorem provable using actual infinity in classical mathematics is provable using operational infinity in VR-Numbers. The two

notions agree on consequences. They differ in ontological status: actual infinity asserts that an infinite totality exists as an object; operational infinity asserts only that an infinite operation is available.

This distinction is significant. The chief philosophical objection to actual infinity, raised since Aristotle and reformulated by intuitionists and constructivists, is the rejection of completed infinities as objects. Operational infinity bypasses this objection: it does not postulate that the totality of natural numbers is a thing. It postulates that the procedure of unbounded counting from $\emptyset$ is available to us, and that any property preserved by this procedure applies universally. The resulting mathematics is the same; the ontological commitment is minimal.

Within VR-Numbers, all references to "infinite" objects — the natural numbers $\mathbb{N}$ as a domain, the integers $\mathbb{Z}$, the rationals $\mathbb{Q}$, the reals $\mathbb{R}$ as Cauchy sequences, the complex numbers $\mathbb{C}$ — are to be understood operationally. None of these collections is an object. Each is a domain of applicability for operations defined over $\emptyset$.

Accordingly, the quantifier notations used throughout this work — expressions of the form " $a, b \in \mathbb{N}$ " or " $\forall \epsilon \in \mathbb{Q_VR}, \epsilon > 0$ " — are operational. They mean: "for any concrete expression of the indicated form". They do not assert the existence of $\mathbb{N}$ or $\mathbb{Q_VR}$ as completed objects. The set-theoretic notation is retained for convenience and conformity with standard mathematical practice; the ontological content is operational throughout.

Note on terminology. Throughout this work, set-theoretic vocabulary ("equivalence class", "set", "field", "subfield") is retained for conformity with standard mathematical practice. Within the operational ontology of VR-Numbers, every such phrase is to be read operationally: an "equivalence class" is the operational rule for recognizing equivalence, not a completed object containing its members; a "field" is the closed system of operations and their results, not a completed collection of values. The forthcoming work *VR-Sets* will provide the operational reconstruction of set-theoretic vocabulary itself. Until then, the standard vocabulary is used with the operational reading understood.

VI.4. The Gradient of Operationality

The structure of VR-Numbers is a gradient: each successive numerical system corresponds to a deeper operational construction over $\emptyset$.

Natural numbers are the most direct operational result: a single operator t is applied repeatedly to the primitive $\emptyset$.

Integers and rationals introduce a layer of formal syntax: expressions over natural numbers, with equivalence relations defining when distinct expressions denote the same operational result.

Real numbers add infinite operational depth: each real number is itself a finite description (algorithm), but the operation it specifies is infinite — producing arbitrarily close rational approximations.

Complex numbers add a second axis: the duality of A1 generates two independent operational directions, and $\mathbb{C}$ is the syntactic structure on these two axes joined by the multiplication rule.

This gradient reflects the historical development of mathematics. Natural numbers have been recognized in all cultures since antiquity, while negative, rational, real, and complex numbers were progressively introduced as the operational needs of mathematics demanded them. The conceptual difficulty historically accompanying each extension — *numeri ficti* for negatives, the irrationality crisis in Greek mathematics, the long dispute over imaginary numbers — reflects, in retrospect, the increasing operational depth of these constructions.

VI.5. What Does Not Exist as an Object

In VR-Numbers, the following are not introduced as ontological objects:

Ordered pairs. Pairs appear nowhere in the construction as independent objects; pairing is replaced by syntactic juxtaposition ($a \ominus b$, $a \oslash b$, $a \oplus b \cdot i$).

Negative numbers. The integer -3 is a notation for the operational class $\{n \ominus (n+3) : n \in \mathbb{N}\}$, not an independent thing.

Fractions. The rational $1/2$ is a notation for the class of equivalent division expressions, not a separate object.

Real numbers. A real number is an equivalence class of Cauchy sequences, which are themselves operational rules — finite specifications of infinite processes — not sets of ordered pairs.

Complex numbers. A complex number is a syntactic expression over two operational axes, not a two-dimensional point in a plane.

This minimalism is the philosophical position of VR-Numbers: only $\oslash$ is. All else is *doing*.

VI.6. Consistency Relative to ZF

Each construction ($\mathbb{Z}_{\text{VR}}$, $\mathbb{Q}_{\text{VR}}$, $\mathbb{C}_{\text{VR}}$) is isomorphic to its standard counterpart. The field of computable real numbers is a definable subfield of $\mathbb{R}$ in ZF (Pour-El & Richards, 1989), and $\mathbb{R}_{\text{VR}}$ is isomorphic to it; therefore $\mathbb{R}_{\text{VR}}$ has a ZF-model. Combined with the standard ZF-models for $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{C}$ and the consistency of VR relative to ZF [Reznik 2026], the full system of VR-Numbers is consistent relative to ZF.

Part VII. Open Questions

(1) Formalization in Type Theory.

The constructions of VR-Numbers — particularly the operational view of functions in the definition of $\mathbb{R}$, and the operational marker i in $\mathbb{C}$ — are naturally suited to constructive type theories such as Martin-Löf's. A formalization in Coq, Agda, or Lean would provide machine verification and explicit categorical semantics.

(2) Algebraic Closure Beyond $\mathbb{C}$.

By the Fundamental Theorem of Algebra, $\mathbb{C}$ is algebraically closed. The theorems of Frobenius (1878) and Hurwitz (1898) further show that $\mathbb{C}$ is the largest finite-dimensional commutative associative normed division algebra over $\mathbb{R}$ — quaternions $\mathbb{H}$ and octonions $\mathbb{O}$ require successive weakening of algebraic properties. Whether these higher-dimensional algebras admit analogous derivations from deeper logical structure — and whether such structure could be located in VR-style logical primitives — remains an open question.

(3) The VR-Sets Program: Operational Set Theory.

A natural extension of the operational ontology is an operational reconstruction of set theory itself, provisionally named *VR-Sets*. The central idea is that the membership relation $x \in y$ admits an operational reading as "x was used in the construction of y", distinguishing *reference* from *physical containment*. Under this reading, the expression $x \in x$ is not paradoxical: it expresses that x is a result of an operation referring to its own name, analogous to recursive structures in programming. This opens a possible unification of well-founded (ZFC-mode) and non-well-founded (ZFA-mode) set theories as two operational modes of a single system, with bisimulation (Aczel, 1988) as the natural equivalence criterion in the ZFA-mode. The technical development of VR-Sets — including the operational reformulation of the ZF axioms, the status of the axiom of choice in operational vs. non-operational regimes, and the relationship between A1's duality and Tarski's theorem on the undefinability of truth — is the subject of forthcoming work.

(4) Computational Realization.

Each level of VR-Numbers admits a direct computational realization, making the system suitable as a foundation for exact-arithmetic computation systems.

Summary

VR-Numbers extends the formal system VR with operational superstructures yielding $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$. The work advances three principal theses:

First, a minimal ontology: the single ontological primitive is the void $\emptyset$; all other objects, including the natural numbers, arise as operations of progressively greater depth performed over $\emptyset$. No ordered pairs are introduced as ontological objects.

Second, actual infinity is replaced by operational infinity: the axiom of induction A4 is read as an operational principle expressing universal applicability of an operation, not as a postulate of a completed infinite totality. The two notions agree on mathematical consequences but differ in ontological commitment.

Third, the two-dimensionality of $\mathbb{C}$ is structurally grounded in the duality of axiom A1: the two generating implications $F \rightarrow F$ and $F \rightarrow \top$ correspond to the real and imaginary axes, respectively. The algebraic coupling of the two axes through $i^2 = \Theta 1$ is an additional joining axiom; A1 motivates the two-dimensionality, while the field structure requires the joining rule. This provides

an ontological motivation for a feature standardly postulated through pairs without deeper justification.

The structures $\mathbb{Z}_{\text{VR}}$, $\mathbb{Q}_{\text{VR}}$, $\mathbb{C}_{\text{VR}}$ are isomorphic to their standard counterparts; $\mathbb{R}_{\text{VR}}$ is isomorphic to the field of computable real numbers (a countable subfield of classical $\mathbb{R}$), reflecting the constructive position of the work. The consistency of the system relative to ZF follows from the consistency of VR established in [Reznik 2026] and from the standard ZF-definability of the computable real numbers (Pour-El & Richards, 1989). The philosophical position is minimal: only $\emptyset$ is — all else is doing.

References

- Aczel, P. (1988). *Non-Well-Founded Sets*. CSLI Lecture Notes 14. Stanford: Center for the Study of Language and Information.
- Bishop, E. (1967). *Foundations of Constructive Analysis*. New York: McGraw-Hill.
- Boolos, G. S., Burgess, J. P., & Jeffrey, R. C. (2007). *Computability and Logic* (5th ed.). Cambridge: Cambridge University Press.
- Cauchy, A.-L. (1821). *Cours d'analyse de l'École royale polytechnique*. Paris: Imprimerie Royale.
- Dedekind, R. (1872). *Stetigkeit und irrationale Zahlen*. Braunschweig: Vieweg.
- Frobenius, G. (1878). Über lineare Substitutionen und bilineare Formen. *Journal für die reine und angewandte Mathematik*, 84, 1–63.
- Hamilton, W. R. (1837). Theory of Conjugate Functions, or Algebraic Couples; with a Preliminary and Elementary Essay on Algebra as the Science of Pure Time. *Transactions of the Royal Irish Academy*, 17, 293–422.
- Hurwitz, A. (1898). Ueber die Composition der quadratischen Formen von beliebig vielen Variablen. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-Physikalische Klasse*, 1898, 309–316.
- Martin-Löf, P. (1984). *Intuitionistic Type Theory*. Naples: Bibliopolis.
- Mendelson, E. (2015). *Introduction to Mathematical Logic* (6th ed.). Boca Raton: Chapman and Hall/CRC.
- von Neumann, J. (1923). Zur Einführung der transfiniten Zahlen. *Acta Litterarum ac Scientiarum*, 1, 199–208.
- Pour-El, M. B., & Richards, J. I. (1989). *Computability in Analysis and Physics*. Berlin: Springer.
- Reznik, V. (2026). VR. A Formal System: A Minimalist Axiomatization of Arithmetic Grounded in Leibnizian Void. Zenodo. DOI: 10.5281/zenodo.20212092.
- Weihrauch, K. (2000). *Computable Analysis: An Introduction*. Berlin: Springer.